Thm. Let E/F be an Extension Field, and let $\alpha, \beta \in E$ be algebraic over F.
α and β are Conjugate if and only if there exists a unique isomorphism $\psi_{\alpha,\beta} : F(\alpha) \to F(\beta)$ which fixes F and maps α to β.

Proof. Suppose that $p(x) \in F[x]$ be the irreducible polynomial having α and β as roots.
Consider the Evaluation Ring Homomorphism,

$$\phi_\alpha : F[x] \to F(\alpha);\ f(x) \mapsto f(\alpha) \quad \text{and} \quad \phi_\beta : F[x] \to F(\beta);\ f(x) \mapsto f(\beta)$$

These have a same kernel, $\ker \phi_\alpha = \ker \phi_\beta = p(x)$. This induces the Natural Iso;

$$\varphi_\alpha : F[x]/(p(x)) \to F(\alpha);\ f(x) + (p(x)) \mapsto f(\alpha)$$

and

$$\varphi_\beta : F[x]/(p(x)) \to F(\beta):\ f(x) + (p(x)) \mapsto f(\beta)$$

Define $\psi_{\alpha,\beta} : F(\alpha) \to F(\beta) : x \mapsto (\varphi_\beta \circ \varphi_\alpha^{-1})(x)$.

Since φ_α^{-1} and φ_β fix F, so is $\psi_{\alpha,\beta}$, and

$$\psi_{\alpha,\beta}(\alpha) = \varphi_\beta(\varphi_\alpha^{-1}(\alpha)) = \varphi_\beta(x + (p(x))) = \beta.$$

So proved. Conversely, observe that

$$\psi_{\alpha,\beta}(\mathrm{irr}(\alpha, F)(\alpha)) = \mathrm{irr}(\alpha, F)(\beta) = 0$$

and $\psi_{\alpha,\beta}^{-1}(\mathrm{irr}(\beta, F)(\beta)) = \mathrm{irr}(\beta, F)(\alpha) = 0$, so both divides each others,
the monicity gives the result.